Chapter 6. Inference about Distributions

Means, paired and two-sample comparisons, variances

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6.1 Inference about a population mean

Motivation. Chapter 5 assumed the population standard deviation σ known, which is rarely the case. Replacing σ by the sample standard deviation S leads to the t distribution and to the t interval and t test for a normal mean.

With σ known we used $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. When σ is unknown it is replaced by its estimator S; this is called **studentizing**. For a random sample from $N(\mu, \sigma^2)$, Theorem 4-10 gives

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Hence

$$P\Big(-t_{\alpha/2}(n-1) \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{\alpha/2}(n-1)\Big) = 1-\alpha, \quad \text{i.e.}$$
$$P\Big(\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}\Big) = 1-\alpha.$$

Definition (Confidence interval for μ (normal population, σ unknown))

A $100(1-\alpha)\%$ confidence interval for μ is

$$\Big(\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}\Big).$$

For large n the normality assumption can be relaxed: by the CLT and since $t(n-1)$ approaches $N(0,1)$, the quantile $t_{\alpha/2}(n-1)$ may be replaced by $z_{\alpha/2}$.

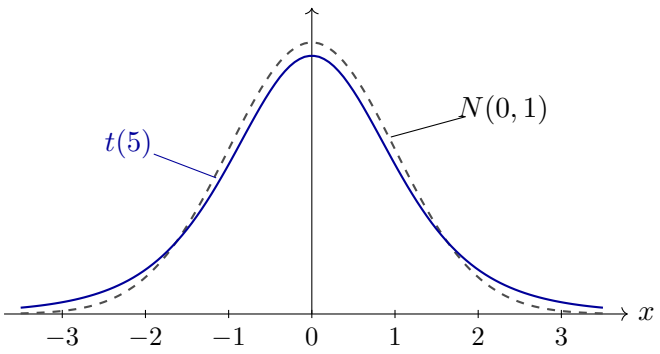


Figure 1: The $t(5)$ density (solid) has heavier tails than $N(0,1)$ (dashed).

Example 6-1 (Trees per 100 m^2)

A forest manager counted trees in 50 randomly chosen 100 m^2 plots of a 9 km^2 forest:

43	28	23	18	46	39	40	43	36	25
26	28	20	22	17	26	17	12	16	35
37	12	32	47	20	25	13	20	46	23
37	29	23	37	34	27	33	29	24	19
37	23	37	28	27	35	30	26	33	35

Find a 99% confidence interval for the mean number μ of trees per 100 m^2.

Solution. With $n = 50$,

$$\bar{x} = \frac{43 + 28 + \cdots + 35}{50} = 28.76, \qquad s = \sqrt{\frac{(43 - 28.76)^2 + \cdots + (35 - 28.76)^2}{49}} = 9.146.$$

The normal quantile plot is close to a line, so a normal population is reasonable. The table lacks 49 degrees of freedom; using the nearest value $t_{0.005}(50) = 2.678$,

$$\bar{x} \pm t_{0.005}(49)\frac{s}{\sqrt{n}} \approx 28.76 \pm 2.678 \times \frac{9.146}{\sqrt{50}} = 28.76 \pm 3.464,$$

so the 99% confidence interval is (25.296, 32.223).

Testing with the studentized mean is the t **test**.

Definition (t test for μ (normal population, σ unknown))

For $H_0 : \mu = \mu_0$ use $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, which is $t(n-1)$ under H_0. With observed value t:

Alternative	P-value	Rejection region at level α
(a) $H_1 : \mu > \mu_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
(b) $H_1 : \mu < \mu_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
(c) $H_1 : \mu \neq \mu_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

(The rejection regions have Type I error exactly α because $T \sim t(n-1)$ when $\mu = \mu_0$; e.g. in (c), $P(|T| \geq t_{\alpha/2}(n-1)) = 2 \cdot \frac{\alpha}{2}$ by symmetry.)

Example 6-2 (Light bulb lifetimes)

Current bulbs last 1950 hours on average. Nine prototypes of a new bulb lasted (hours)

2000	1975	1900	2000	1950	1850	1950	2100	1975

(a) State suitable hypotheses.
(b) Assuming normal lifetimes, test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 1950$, $H_1 : \mu > 1950$.
(b) $\bar{x} = 1966.7$, $s = 69.6$, so $t = \dfrac{1966.7 - 1950}{69.6/\sqrt{9}} = 0.718$. The rejection region is $t \geq t_{0.05}(8) = 1.860$; since $0.718 < 1.860$, H_0 is not rejected at 5%.
(c) $P = P(T \geq 0.718)$, $T \sim t(8)$. From the table, 0.718 lies between the upper 25% point (0.706) and 20% point (0.889), so $0.20 < P < 0.25$; software gives $P = 0.247$. The evidence that the new bulb lasts longer is quite weak.

The normal quantile plot of these data looks roughly linear, but with so few observations such a check has limited power; one may instead rely on subject knowledge, or use the nonparametric methods of Chapter 10.

Example 6-3 (Cholesterol of immigrant women)

The national mean cholesterol for women is 190 mg/dL. For 100 randomly selected women who immigrated in the past year, $\bar{x} = 178.2$ and $s = 45.3$.
(a) State suitable hypotheses.
(b) Test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 190$, $H_1 : \mu \neq 190$.
(b) $t = \dfrac{178.2 - 190}{45.3/\sqrt{100}} = -2.60$. With $n = 100$ the t test applies without assuming normality, and $z_{0.025} = 1.96$ can replace the $t(99)$ quantile: the rejection region is $|t| \geq 1.96$. Since $|t| = 2.60 \geq 1.96$, H_0 is rejected at 5%.
(c) $P = P(|T| \geq 2.60) \approx 2 \times 0.0047 = 0.0094$ from the normal table; software gives $P = 0.0107$. The mean cholesterol of immigrant women differs significantly from the national mean, so follow-up measures (lifestyle changes, medication) may be recommended.

6.2 Comparing means by paired comparison

Motivation. Many questions compare two populations or treatments. When the units are heterogeneous, applying both treatments to each unit (or to matched pairs) removes the unit-to-unit variation, and the comparison reduces to a one-sample problem for the differences.

Comparing two means by observing pairs and analysing the within-pair differences is called **paired comparison**.

Illustration 6-1: two sleeping pills

Patients' health varies greatly, so each of 6 randomly chosen patients took both pills A and B once, in an order decided by a coin toss, and sleeping hours were recorded:

Patient	1	2	3	4	5	6
Pill A	4.8	4.0	5.8	4.9	5.3	7.4
Pill B	4.0	4.2	5.2	4.9	5.6	7.1

What can be inferred about $\mu_A - \mu_B$?

Illustration 6-2: wearable walking robot

To see whether a gait-assist robot increases exercise effect, 15 rehabilitation patients each walked 10 minutes with and without the robot; calories burned (kcal):

Patient	1	2	3	4	5	6	7	8
Without	23.9	23.9	20.0	19.2	22.2	11.8	21.7	27.7
With	25.2	28.2	24.9	20.2	19.7	16.9	22.6	20.8
Patient	9	10	11	12	13	14	15	
Without	15.6	22.5	23.2	19.3	15.2	10.0	11.2	
With	23.8	18.3	23.7	22.5	16.2	12.1	22.5	

In both illustrations the objects receiving the treatments (patients) are **experimental units**; the factors applied (pills, robot) are **treatments**; the change they produce is the **treatment**

effect. Pairing homogeneous units sharpens the comparison of two treatments. With data $(X_1, Y_1), \ldots, (X_n, Y_n)$, $\mu_1 = E(X_i)$, $\mu_2 = E(Y_i)$, form

$$D_i = X_i - Y_i, \qquad E(D_i) = \mu_1 - \mu_2, \qquad i = 1, \ldots, n,$$

and apply §6.1 to the sample $D_1, \ldots, D_n$ with mean $\bar{D}$ and standard deviation S_D.

Definition (Paired comparison of two means)

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\Big(\bar{d} - t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}},\ \bar{d} + t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}\Big).$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ the test statistic is $T = \dfrac{\bar{D} - \delta_0}{S_D/\sqrt{n}}$; with observed value t:

Alternative	P-value	Rejection region at level α
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

These are exact when $D_1, \ldots, D_n$ is a random sample from a normal distribution (Theorem 4-10 applied to the D_i), and approximate for large n. To meet the assumptions, the experimental units should be selected at random and the order of treatments randomized (e.g. by a coin toss): **randomization**.

Example 6-4 (Sleeping pills: confidence interval)

Using Illustration 6-1, find a 95% confidence interval for $\mu_A - \mu_B$.
Solution. The differences A−B are 0.8, −0.2, 0.6, 0.0, −0.3, 0.3, so $\bar{d} = 0.20$ and $s_D = 0.443$. With $t_{0.025}(5) = 2.571$,

$$\bar{d} \pm t_{0.025}(5)\frac{s_D}{\sqrt{n}} = 0.20 \pm 2.571 \times \frac{0.443}{\sqrt{6}} = 0.20 \pm 0.46,$$

so the 95% confidence interval is $(-0.26, 0.66)$.

A normal quantile plot can check normality here, but more important is randomization in planning and observation.

Example 6-5 (Walking robot: test)

Using Illustration 6-2:
(a) test at level 1% whether wearing the robot increases calorie consumption;
(b) find the P-value of (a).
Solution.
(a) With μ_1, μ_2 the mean calories without and with the robot, $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$. The $n = 15$ differences (without − with) are

$$-1.3,\ -4.3,\ -4.9,\ -1.0,\ 2.5,\ -5.1,\ -0.9,\ 6.9,\ -8.2,\ 4.2,\ -0.5,\ -3.2,\ -1.0,\ -2.1,\ -11.3,$$

with $\bar{d} = -2.01$ and $s_D = 4.58$, so $t = \dfrac{-2.01 - 0}{4.58/\sqrt{15}} = -1.70$. Since $t_{0.01}(14) = 2.624$, the rejection region is $t \leq -2.624$; as $-1.70 > -2.624$, H_0 is not rejected at 1%.
(b) $P = P(T \leq -1.70)$, $T \sim t(14)$: the table gives $0.05 < P < 0.1$, and software $P = 0.056$.

At the 1% level there is no clear evidence that the robot increases calorie consumption.

A normal quantile plot of these differences supports the normality assumption.

6.3 Comparing means with two independent samples

Motivation. When the same unit cannot receive both treatments, the units are split into two groups (or two populations are sampled separately). Inference about $\mu_1 - \mu_2$ then rests on the difference of two independent sample means.

Definition (Data structure for two-sample comparison)

(a) The observations in each group are a random sample from that group's population.
(b) Observations in different groups are independent.

Random samples from two populations satisfy this automatically; when one pool of units receives two treatments, the units must be split into the groups at random (randomization).

Label the populations $1, 2$ with means μ_1, μ_2 and standard deviations σ_1, σ_2, and let $X_{11}, \dots, X_{1n_1}$ and $X_{21}, \dots, X_{2n_2}$ be the samples, with

$$\bar{X}_1 = \frac{X_{11} + \cdots + X_{1n_1}}{n_1}, \quad S_1 = \sqrt{\frac{1}{n_1 - 1}\sum_{j=1}^{n_1}(X_{1j} - \bar{X}_1)^2},$$

and $\bar{X}_2, S_2$ likewise. We estimate $\mu_1 - \mu_2$ by $\widehat{\mu_1 - \mu_2} = \bar{X}_1 - \bar{X}_2$. By independence,

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \text{so} \qquad \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

is the standardized difference, which is exactly $N(0,1)$ for normal populations (Theorem 4-2). With σ_1^2, σ_2^2 unknown we use the studentized difference

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}.$$

6.3.1 Large samples

When n_1, n_2 are both large, $S_1^2 \approx \sigma_1^2$ and $S_2^2 \approx \sigma_2^2$, so T is approximately $N(0,1)$ by the CLT.

Definition (Two-sample comparison of means (large n_1, n_2))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$, $T \dot{\sim} N(0,1)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq z_\alpha$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -z_\alpha$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq z_{\alpha/2}$

Example 6-6 (Tennis rackets and tennis elbow)

A new racket is designed to reduce the stress on the elbow. Of 220 intermediate players, 100 were randomly assigned the new racket and 120 the old one; the force on the elbow at impact in a forehand stroke (kgf) was:

	n_i	$\bar{x}_i$	s_i
New racket	100	25.2	5.1
Old racket	120	27.9	5.6

(a) Test at level 5% whether the mean forces differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. The samples are large, and

$$t = \frac{(25.2 - 27.9) - 0}{\sqrt{5.1^2/100 + 5.6^2/120}} = -3.74.$$

The rejection region is $|t| \geq z_{0.025} = 1.96$; since $3.74 \geq 1.96$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 3.74)$ with $T \dot{\sim} N(0,1)$: $P < 0.0006$ from the table, and $P = 0.00018$ by software.
(c) $(25.2 - 27.9) \pm 1.96\sqrt{5.1^2/100 + 5.6^2/120} = -2.7 \pm 1.42$, i.e. $(-4.12, -1.28)$.

6.3.2 Normal populations: Welch's approximation

Even when n_1, n_2 are not large, T is approximately a t distribution with suitably chosen degrees of freedom. For normal populations with $n_1, n_2 \geq 5$,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df), \qquad df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2}.$$

Definition (Two-sample comparison of means (normal populations))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(df)\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

with df computed from s_1^2, s_2^2 by the formula above.
(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(df)$

The normality assumption matters little once n_1, n_2 are moderately large, so these methods may be used unless the Q–Q plots strongly contradict normality. Since df is generally not an integer, use the nearest tabulated value, interpolate, or use software.

Example 6-7 (Grocery spending offline vs. online)

An economic psychologist claims that grocery spending is lower offline than online. Randomly chosen 18 and 20 people shopped two weeks' groceries for a family of four offline and online, respectively (thousand won):

Offline	169	206	257	294	252	283	240	207	230	183
	298	269	256	277	300	126	318	184		
Online	289	334	278	268	336	438	388	388	394	394
	425	386	356	342	305	365	355	312	209	458

(a) Test at level 5% whether offline spending is lower.
(b) Find the P-value of (a).
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$ (1 = offline, 2 = online). Summary: $n_1 = 18$, $\bar{x}_1 = 241.6$, $s_1 = 52.82$; $n_2 = 20$, $\bar{x}_2 = 351.0$, $s_2 = 61.92$. The Q–Q plots of both groups are close to lines. Then

$$t = \frac{(241.6 - 351.0) - 0}{\sqrt{52.82^2/18 + 61.92^2/20}} = -5.875, \qquad df = \frac{\left(\frac{52.82^2}{18} + \frac{61.92^2}{20}\right)^2}{\frac{1}{17}\left(\frac{52.82^2}{18}\right)^2 + \frac{1}{19}\left(\frac{61.92^2}{20}\right)^2} = 35.91.$$

The rejection region is $t \leq -t_{0.05}(35.91)$, where $t_{0.05}(35.91) \approx 1.620$ by interpolation in the table (1.688 by software). In either case $t = -5.875$ lies in the rejection region, so H_0 is rejected at 5%.
(b) $P = P(T \leq -5.875)$, $T \sim t(35.91)$; the table gives $P < 0.0005$. There is clear evidence that offline grocery spending is lower.

6.3.3 Normal populations with a common variance

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, both S_1^2 and S_2^2 estimate σ^2, and combining them is more efficient. Weighting by degrees of freedom gives the **pooled sample variance**

$$S_p^2 = \frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} S_1^2 + \frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} S_2^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

with $n_1 + n_2 - 2$ degrees of freedom (Theorem 4-9).

	Population 1	Population 2	Pooled
Variance	σ^2	σ^2	
Sample variance	S_1^2	S_2^2	S_p^2
Degrees of freedom	$n_1 - 1$	$n_2 - 1$	$n_1 + n_2 - 2$

Under a common variance the standardized difference is $\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+1/n_2}}$; replacing σ by S_p,

$$T=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}\sim t(n_1+n_2-2)$$

for normal populations, by Theorem 4-11.

Definition (Two-sample comparison (normal populations, common variance))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1-\mu_2$ is

$$(\bar{x}_1-\bar{x}_2)\pm t_{\alpha/2}(n_1+n_2-2)\,s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}.$$

(b) For $H_0:\mu_1-\mu_2=\delta_0$ use $T=\dfrac{(\bar{X}_1-\bar{X}_2)-\delta_0}{S_p\sqrt{1/n_1+1/n_2}}$; with observed value t:

Alternative	P-value	Rejection region
$H_1:\mu_1-\mu_2>\delta_0$	$P(T\geq t)$	$t\geq t_\alpha(n_1+n_2-2)$
$H_1:\mu_1-\mu_2<\delta_0$	$P(T\leq t)$	$t\leq -t_\alpha(n_1+n_2-2)$
$H_1:\mu_1-\mu_2\neq\delta_0$	$P(\lvert T\rvert\geq\lvert t\rvert)$	$\lvert t\rvert\geq t_{\alpha/2}(n_1+n_2-2)$

This is the **pooled two-sample t test**; the previous one, with adjusted degrees of freedom and no common-variance assumption, is the **unpooled two-sample t test**.

Pooled or unpooled?

The pooled test requires equal variances, and its validity deteriorates badly if they differ, especially when n_1 and n_2 are very different. Equal variances are hard to verify from box plots or histograms, so the unpooled test should also be examined; use the pooled test only when confident of a common variance.

Example 6-8 (Semaglutide and weight loss)

Twelve similar volunteers were split at random into two groups of 6: one received weekly semaglutide injections, the other a placebo, for 12 weeks with diet and exercise controlled. Weight loss (kg):

Semaglutide	5.12	4.87	5.64	5.01	5.49	5.25
Placebo	3.62	3.98	4.14	3.25	4.42	3.60

(a) Test at level 5% whether the mean weight losses differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.

Solution.

(a) $H_0:\mu_1=\mu_2$, $H_1:\mu_1\neq\mu_2$. Summary: $n_1=n_2=6$, $\bar{x}_1=5.23$, $s_1=0.29$; $\bar{x}_2=3.84$, $s_2=0.42$. The Q–Q plots support normality and the standard deviations are similar, so we assume a common variance:

$$s_p^2=\frac{(6-1)\times 0.29^2+(6-1)\times 0.42^2}{6+6-2}=0.13025,\qquad s_p=\sqrt{0.13025}=0.361,$$

$$t=\frac{(5.23-3.84)-0}{0.361\sqrt{\frac{1}{6}+\frac{1}{6}}}=6.67.$$

The rejection region is $|t| \geq t_{0.025}(10) = 2.228$; since $6.67 > 2.228$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 6.67)$, $T \sim t(10)$: $P < 0.001$. There is strong evidence that the mean weight losses differ.
(c) $(5.23 - 3.84) \pm 2.228 \times 0.361 \times \sqrt{\frac{1}{6} + \frac{1}{6}} = 1.39 \pm 0.46$, i.e. (0.93, 1.85).

The two-sample t tests and intervals are not very sensitive to the normality assumption and are approximately valid for large samples. What really matters is that the experiment was randomized.

6.4 Inference about population variances

Motivation. Variability itself is often the target: a process producing parts of the right average size but with large variation still produces many defects. For normal populations, inference on σ^2 uses the chi-square distribution and comparison of two variances uses the F distribution.

6.4.1 One variance

For a random sample of size n from $N(\mu, \sigma^2)$ with μ, σ unknown, Theorem 4-8 gives

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1).$$

With the quantiles $\chi^2_{1-\alpha/2}(n-1)$ and $\chi^2_{\alpha/2}(n-1)$,

$$P\Big(\chi^2_{1-\alpha/2}(n-1) \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2}(n-1)\Big) = 1-\alpha,$$

and solving the inequalities for σ^2,

$$P\Big((n-1)\frac{S^2}{\chi^2_{\alpha/2}(n-1)} \leq \sigma^2 \leq (n-1)\frac{S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) = 1-\alpha.$$

Definition (Confidence interval for σ^2 (normal population))

A $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\Big((n-1)\frac{s^2}{\chi^2_{\alpha/2}(n-1)},\ (n-1)\frac{s^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

The test of a variance based on the chi-square distribution is the **chi-square test**.

Definition (Chi-square test for σ^2 (normal population))

For $H_0 : \sigma^2 = \sigma_0^2$ use $\chi^2 = (n-1)\frac{S^2}{\sigma_0^2}$, which is $\chi^2(n-1)$ under H_0. With observed value χ_0^2:

Alternative	P-value	Rejection region at level α
$H_1 : \sigma^2 > \sigma_0^2$	$P(\chi^2 \geq \chi_0^2)$	$\chi_0^2 \geq \chi^2_\alpha(n-1)$
$H_1 : \sigma^2 < \sigma_0^2$	$P(\chi^2 \leq \chi_0^2)$	$\chi_0^2 \leq \chi^2_{1-\alpha}(n-1)$
$H_1 : \sigma^2 \neq \sigma_0^2$	$2P(\chi^2 \geq \chi_0^2)$ or $2P(\chi^2 \leq \chi_0^2)$	$\chi_0^2 \leq \chi^2_{1-\alpha/2}(n-1)$ or $\chi_0^2 \geq \chi^2_{\alpha/2}(n-1)$

For the two-sided alternative, compute both $2P(\chi^2 \geq \chi_0^2)$ and $2P(\chi^2 \leq \chi_0^2)$ and use the one smaller than 1.

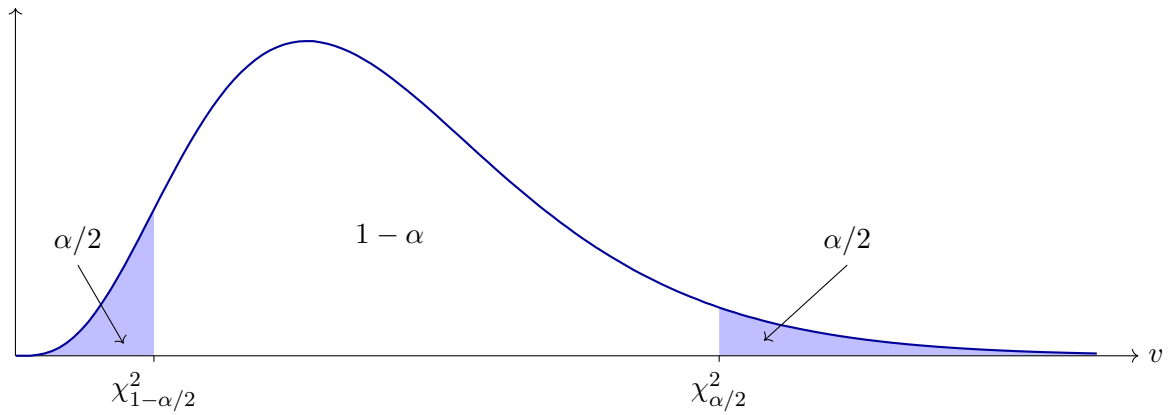


Figure 2: Quantiles of $\chi^2(9)$ used for a 90% interval for σ^2 ($\alpha = 0.1$): $\chi^2_{0.95}(9) = 3.33$, $\chi^2_{0.05}(9) = 16.92$.

Example 6-9 (Coffee grinder)

A café uses 10 g of beans per Americano, but the grinder dose varies; a standard deviation above 1.5 g is considered abnormal. Ten doses (g):

11.15	9.62	9.83	10.85	12.66	9.51	11.23	9.38	9.11	8.56

Past records indicate a normal distribution.
(a) Test at level 5% whether the grinder is abnormal.
(b) Find the P-value of (a).
(c) Find a 90% confidence interval for the standard deviation of the dose.
Solution.
(a) $H_0 : \sigma^2 = 1.5^2$, $H_1 : \sigma^2 > 1.5^2$. Here $\bar{x} = 10.19$, $s = 1.24$, and

$$\chi_0^2 = (n-1)\frac{s^2}{\sigma_0^2} = 9 \times \frac{1.24^2}{1.5^2} = 6.15.$$

The rejection region is $\chi_0^2 \geq \chi^2_{0.05}(9) = 16.92$; since $6.15 < 16.92$, H_0 is not rejected at 5%.
(b) $P = P(\chi^2 \geq 6.15)$, $\chi^2 \sim \chi^2(9)$; software gives $P = 0.72$. There is no clear evidence of abnormal variation.
(c) With $\chi^2_{0.05}(9) = 16.92$ and $\chi^2_{0.95}(9) = 3.33$, the 90% interval for σ^2 is

$$\left(9 \times \frac{1.24^2}{16.92},\ 9 \times \frac{1.24^2}{3.33}\right) = (0.818,\ 4.156),$$

so the interval for σ is $(\sqrt{0.818}, \sqrt{4.156}) = (0.90,\ 2.04)$.

Sensitivity to normality

Unlike the t test and t interval for a mean, chi-square inference on a variance depends heavily on the normality assumption; check it (e.g. by a Q–Q plot, which is roughly linear for the data of Example 6-9) before using these methods.

6.4.2 Ratio of two variances

For independent random samples $X_{11},\dots,X_{1n_1}$ and $X_{21},\dots,X_{2n_2}$ from normal populations, with sample variances S_1^2, S_2^2, Theorem 4-13 gives

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1).$$

Therefore

$$P\Big(F_{1-\alpha/2}(n_1-1,n_2-1) \le \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \le F_{\alpha/2}(n_1-1,n_2-1)\Big) = 1-\alpha,$$

and solving for σ_1^2/σ_2^2,

$$P\Big(\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}\Big) = 1-\alpha.$$

By Theorem 4-12, $F_{1-\alpha/2}(n_1-1,n_2-1) = 1/F_{\alpha/2}(n_2-1,n_1-1)$.

Definition (Confidence interval for σ_1^2/σ_2^2 (normal populations))

A $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{s_1^2}{s_2^2}\times\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)},\ \frac{s_1^2}{s_2^2}\times F_{\alpha/2}(n_2-1,n_1-1)\Big).$$

The test of σ_1^2/σ_2^2 based on the F distribution is the F **test**.

Definition (F test for the ratio of variances (normal populations))

For $H_0:\sigma_1^2=\sigma_2^2$ use $F=S_1^2/S_2^2$, which is $F(n_1-1,n_2-1)$ under H_0. With observed value f:

Alternative	P-value	Rejection region at level α
$H_1:\sigma_1^2>\sigma_2^2$	$P(F\ge f)$	$f \ge F_{\alpha}(n_1-1,n_2-1)$
$H_1:\sigma_1^2<\sigma_2^2$	$P(F\le f)$	$f \le 1/F_{\alpha}(n_2-1,n_1-1)$
$H_1:\sigma_1^2\ne\sigma_2^2$	$2P(F\ge f)$ or $2P(F\le f)$	$f \le 1/F_{\alpha/2}(n_2-1,n_1-1)$ or $f \ge F_{\alpha/2}(n_1-1,n_2-1)$

Example 6-10 (Polymer injection pressure)

Cracks in concrete are repaired by injecting a polymer. Injection pressures (10^6 psi) for two polymers:

Epoxy	1.75	2.12	2.05	1.97
MMA prepolymer	1.77	1.59	1.70	1.69

(a) Test at level 5% whether the variabilities of the injection pressures differ.
(b) Find a 95% confidence interval for the ratio of standard deviations.
Solution.
(a) $H_0:\sigma_1^2=\sigma_2^2$, $H_1:\sigma_1^2\ne\sigma_2^2$. Summary: $n_1=n_2=4$, $\bar{x}_1=1.97$, $s_1=0.160$; $\bar{x}_2=1.69$, $s_2=0.074$. The Q–Q plots do not contradict normality. The observed statistic is $f=0.160^2/0.074^2=4.675$. Since $F_{0.025}(3,3)=15.44$, the 5% rejection region is $f\le 1/15.44$ or $f\ge 15.44$. As $1/15.44<4.675<15.44$, H_0 is not rejected at 5%.

(b) The 95% interval for σ_1^2/σ_2^2 is

$$\Big(\frac{0.160^2}{0.074^2} \times \frac{1}{15.44},\ \frac{0.160^2}{0.074^2} \times 15.44\Big) = (0.301,\ 72.182),$$

so that for σ_1/σ_2 is $(\sqrt{0.301}, \sqrt{72.182}) = (0.549,\ 8.496)$.